

Hydraulic Cylinder

A hydraulic cylinder is a mechanical actuator that converts **hydraulic energy into linear motion and force**. It is widely used in industrial, mobile, and construction machinery.

Types of Hydraulic Cylinders

1) Single Acting Hydraulic Cylinder

- Hydraulic pressure acts in **one direction only**
- Return stroke is achieved by **spring force, gravity, or external load**
- Simple construction and lower cost
- Common applications:
 - Lifting equipment
 - Presses
 - Dump bodies

2) Double Acting Hydraulic Cylinder

- Hydraulic pressure acts in **both extension and retraction**
- Provides controlled motion in both directions
- Higher efficiency and precision
- Common applications:
 - Machine tools
 - Construction equipment
 - Automation systems

Cylinder Construction Types

A) Welded Type Hydraulic Cylinder

In welded cylinders, the barrel is **welded directly to the end caps**, making them compact and robust.

Mounting Options:

1. Front Flange



A front flange welded cylinder is a heavy-duty hydraulic cylinder with a strong, welded body and a flat mounting plate (flange) attached to its rod end, allowing for secure, rigid attachment to machinery like trucks, cranes, and industrial equipment for lifting, steering, and pushing tasks, offering durability and stability.

2. Rear Flange



A rear flange welded cylinder is a heavy-duty hydraulic cylinder with a sturdy, welded mounting plate (flange) attached to its back end, providing stable, fixed-point installation for heavy machinery, offering strength, precise alignment, and durability for lifting, pressing, and industrial automation tasks.

3. Rear Clevis



A **Rear Clevis Welded Cylinder** is a robust hydraulic cylinder featuring a U-shaped mounting bracket (clevis) permanently welded to its rear end, designed for heavy-duty applications requiring strong pivot points, common in machinery like trailers, agricultural equipment, and industrial presses for lifting, pushing, and pulling loads with stability and durability.

4. Both End Clevis



A "Both End Clevis welded cylinder" is a type of industrial hydraulic or pneumatic cylinder with a U-shaped mounting bracket (clevis) welded onto *both* the rod end and the base end, allowing it to pivot and connect to machinery in two places for applications needing movement in an arc, common in construction, agriculture, and lifting equipment. These cylinders are robust, often double-acting (using fluid pressure for both extension and retraction), and built for heavy-duty use in tasks like pushing, pulling, and lifting loads.

5. Rod End Clevis

A **ROD End Clevis Welded Cylinder** is a heavy-duty hydraulic cylinder where the clevis (a U-shaped mounting bracket) is permanently welded to the piston rod's end, creating a robust, compact unit ideal for rugged environments like agriculture, construction, and industrial machinery, allowing for angular movement and connecting to machine linkages securely.

6. Center Trunnion



A **center Trunnion mounting welded cylinder** is a heavy-duty hydraulic cylinder with pivot pins (Trunnion) welded directly onto its body, usually at the center, allowing it to swing or pivot like a hinge for controlled linear motion, ideal for applications needing side-to-side movement or self-alignment, often built with robust welded construction for high pressures in industrial settings like heavy machinery.

7. Front Trunnion



A **Front Trunnion Mounting Welded Cylinder** is a heavy-duty hydraulic cylinder with pivot pins (Trunnion) attached to the front end, allowing it to swing for flexible, controlled motion in applications like heavy machinery (dozers, cranes) and industrial automation, featuring a robust, leak-resistant design with barrel and end caps directly welded together for durability, often made of mild steel.

8. Rear Trunnion



A **rear Trunnion mounting welded cylinder** is a heavy-duty hydraulic cylinder with pivot pins (Trunnion) attached near the rod end (rear) of the barrel, allowing it to swing or pivot for smoother movement, often used in machinery needing angular motion and high load support, with its barrel constructed by welding components like the end caps to the main tube for robust, leak-free operation.

B) Tie Rod Type Hydraulic Cylinder

Tie rod cylinders use **high-strength tie rods** to hold the end caps to the barrel.

Mounting Options:

- 1. Front Flange



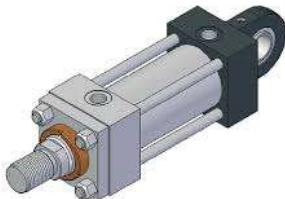
A **Tie Rod Type Hydraulic Cylinder with a Front Flange** is a heavy-duty linear actuator using steel tie rods for strength, featuring a flat mounting plate (flange) on the head (rod) end for direct attachment to machinery, ideal for pushing/pulling in industrial settings, known for durability, easy maintenance, and common in applications needing robust, fixed-position mounting like presses or lifts, with variations for pressure (e.g., 200-300 Bar), bore size, and stroke length.

- 2. Rear Flange



A **Tie Rod Type Hydraulic Cylinder with a Rear Flange** is a durable, common industrial cylinder where steel rods (tie rods) run the length of the cylinder, securing end caps, and features a flange at the back (cap end) for rigid mounting onto machinery, ideal for thrust/compression loads, offering easy maintenance and various customization options for automation and construction.

- 3. Rear Clevis



A [tie rod hydraulic cylinder](#) with a rear clevis mounting uses a sturdy clevis at the cylinder's base (cap end) to pivot and connect to machinery, allowing for angular movement, common in industrial tasks needing robust, serviceable cylinders with replaceable parts, often conforming to standards like ISO 6020/2, and featuring high-strength rods, precision-honed tubes, and durable seals for reliable linear force in heavy-duty applications.

- 4. Both End Clevis



A **Tie Rod Type Hydraulic Cylinder with Both End Clevis** is a robust, repairable cylinder using threaded rods (tie rods) to secure end caps, featuring U-shaped clevis mounts on *both* ends for versatile, pivoting connections (allowing arc motion), ideal for heavy industrial, agricultural, or construction machinery needing strength, field serviceability, and secure, adaptable mounting to handle push/pull forces.

- 5. Rod End Clevis

A [tie-rod hydraulic cylinder](#) with a clevis rod end features a U-shaped fork (clevis) at the rod end, often secured by a pin, allowing the cylinder to pivot and articulate in one plane, ideal for mobile/construction use where alignment changes are needed; these cylinders use long tie-rods to hold end caps, offering easy field repair and modularity, commonly meeting NFPA standards, and provide linear force in both extension and retraction for heavy-duty lifting and pressing applications.

6. Center Trunnion



A **Tie Rod Type Hydraulic Cylinder with [Center Trunnion](#)** combines the strong, serviceable design of tie-rod construction (long bolts holding end caps) with a **center Trunnion** for pivot mounting, allowing it to rotate around its middle axis for applications needing controlled pivoting, making it durable, easily repairable, and suited for heavy industrial use where alignment and movement flexibility are crucial, unlike fixed-end mounts

7. Front Trunnion



A **Tie Rod Type Hydraulic Cylinder with [Front Trunnion](#)** combines the durable, easily-maintained design of tie-rod cylinders (long rods securing end caps) with a Trunnion mount at the rod (front) end, allowing for pivotal, swinging motion, ideal for applications needing guided linear movement that pivots from the front, like lifting equipment or presses, offering high force and reliable reparability in industrial settings.

8. Rear Trunnion

A **Tie Rod Type Hydraulic Cylinder with Rear Trunnion** combines a durable, tie-rod-secured body with a pivot point (Trunnion) at the back end, allowing it to swing or pivot while generating linear force, ideal for heavy industrial tasks like construction, material handling, and presses where angular movement and robust construction are needed for stability and long life.

9. Foot Mounting



A **Tie Rod Type Hydraulic Cylinder with Foot Mounting** is a robust, industrial hydraulic actuator featuring threaded rods (tie rods) securing end caps, designed to be bolted directly onto a flat surface (foot mount) for stable, stationary linear motion in applications like presses, lifts, and machinery. These cylinders are known for their durability, high pressure handling, and common use in heavy industrial settings, offering excellent load distribution and alignment with their base support.